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RESEARCH ARTICLE



A streamlined approach to uncertainty and sensitivity analysis for models with spatial outputs with an example from geodiversity assessment

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ABSTRACT

The paper proposes a joint use of two global sensitivity analysis methods to evaluate the influence of model inputs on spatially explicit outputs. The analysis is conducted on spatial outputs of geodiversity assessment. First, the screening method is used to identify inputs with low importance and interactions, which can be discarded in later analysis. A simplified model is compared with a complete model using the mean and variance calculated on spatial outputs to evaluate outcome uncertainty. Finally, the variance decomposition method computes two sensitivity index maps per input. The results are mapped using bivariate classification, allowing for the simultaneous identification of regions with higher geodiversity scores and their corresponding variability. The research reveals that the streamlined approach preserves the overall patterns of outcome uncertainty and sensitivity observed in an original model while simultaneously reducing the number of outcome maps to interpret. Additionally, bivariate mapping proves efficient in concurrently depicting spatial sensitivity and interaction effects. The analytic efficiencies afforded by the presented approach and bivariate mapping enhance the operational knowledge of spatially explicit uncertainty and sensitivity analysis by simplifying the computational procedure and offering an alternative way of visualizing and interpreting input factors driving the sensitivity of model output.

KEY POLICY HIGHLIGHTS

- Spatially explicit uncertainty and sensitivity analysis extends the geodiversity assessment methods and comprises the next step in improving the accuracy of geodiversity assessments for land management decisions.
- An integrated uncertainty and sensitivity analysis helps focus on high-priority areas (high sensitivity and low uncertainty) to improve assessment reliability.

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- The uncertainty and sensitivity analysis in models with spatially explicit output results depicted on maps can help reduce field research expenses by identifying a particular area or a specific natural component that warrants better-quality data.
- Bivariate maps employed to visualize the results of spatially explicit uncertainty and sensitivity analysis can potentially reduce the cognitive effort of visually switching between multiple single-variable (theme) maps.

1. Introduction

Uncertainty is a pervasive element of spatial modeling, present in model inputs and their formulations. It manifests itself in the conception of the model, data collection, model evaluation, and its application. For example, inputs obtained from experts can exhibit variations in values, leading to an input distribution instead of a constant value. Furthermore, translating abstract concepts into quantifiable variables can be subjective (eg a model requiring an input reflecting the objective of freshwater availability can be operationalized as distance to surface water, access to wells, or the quality of water infrastructure). The presence of uncertain model components introduces variability into outcome values, such as generating multiple alternative output maps. This complexity impacts the synthesis, interpretation, and generalization of results, as well as the model's transferability to different study areas and the establishment of stakeholder trust in the model. Consequently, handling model uncertainty emerges as a critical aspect of model evaluation.

Uncertainty and sensitivity analysis (US-A) has long been an established method of testing the reliability of model predictions (Saltelli et al. 2008, Ferretti et al. 2016, Razavi et al. 2021). The two-phase method starts with systematically sampling model inputs and examining the effects on the model's output variability (phase one comprising uncertainty analysis) and follows with attributing the variability to specific inputs (phase two comprising sensitivity analysis). The uncertainty and sensitivity must be analyzed in tandem to comprehensively evaluate model input/output uncertainty. Given the variable input values, uncertainty analysis is necessary to generate the output distribution. Then, sensitivity analysis informs us about which of those inputs drive the output variability, identifying the influential inputs and leading to model simplification by fixing the values of variables with little to no effect on the output variability.

US-A has been applied in spatial analysis and modeling (Gómez-Delgado and Tarantola 2006, Marrel et al. 2011, Chen et al. 2013, Herman et al. 2013a, Ligmann-Zielinska 2013, Moreau et al. 2013, Xu and Zhang 2013, Feizizadeh et al. 2014, Ligmann-Zielinska et al. 2014, Şalap-Ayça et al. 2018, Kang et al. 2020, Koo et al. 2020, Roux et al. 2021). The seminal paper by Lilburne and Tarantola (2009) provided one of the earliest discussions on alternative approaches to performing US-A in a spatial setting, promoting global approaches that simultaneously probe the whole input space and evaluate how the simultaneous changes in all inputs affect the variability of outputs both individually and in combinations. In contrast to the global approaches, a one-at-a-time or OAT approach probes the input space locally by varying one input at

a time while keeping the other inputs unchanged. While attractive due to its simplicity (Chen et al. 2013), OAT has proven to be inefficient or even deceptive when dealing with many interacting inputs (Saltelli and Annoni 2010), which is characteristic of complex models, including spatial models.

Most applications in the context of US-A have focused on models where the analyzed output variables are conceptualized as spatially invariant, including agent-based models and multicriteria decision-making (MCDM) models. Examples of agent-based models include the total predicted panda habitat area (An et al. 2005), land enrolled into a conservation program (Ligmann-Zielinska et al. 2014), or evaluation of land markets (Huang et al. 2013).

Within the realm of MCDM, both the OAT and global sensitivity analysis approaches have found applications (Malczewski and Rinner 2015). The OAT approach limits the search for sensitive inputs to a small input value space. The approach has been popular in GIS-MCDM applications, particularly in the 1990s and 2000s (Jankowski et al. 1997, Feick and Hall 2004, Chen et al. 2013). In a 2004 review, Gomez Degaldo and Bosque-Sendra observed that sensitivity analysis was not being employed to pinpoint the drivers of model uncertainty in MCDM. Instead, it was primarily utilized to assess the model's stability, where the OAT approach focused on criterion weights. Ligmann-Zielinska and Jankowski (2008) pointed out in their review that applications of US-A in GIS-MCDM utilized aspatial US-A methods developed in statistics, engineering, and management science, but not in GIScience. They suggested developing spatially explicit methods for US-A to better capture the model's sensitivity to input uncertainty, considering spatial heterogeneities.

Most global uncertainty and sensitivity analyses of MCDM utilize variance decomposition (Saltelli et al. 2008). In an early study employing the global approach, Gómez-Delgado and Tarantola (2006) utilized global variance decomposition to assess the stability of a model's solutions when confronted with uncertainties affecting both criteria values and weights. Their study applied sensitivity analysis to aggregated criteria values and weights, yielding a single sensitivity measure for each input within the study area. While capable of identifying influential inputs driving the MCDM model's output variability, this spatially implicit approach crucially lacked spatial representation of these influences. A spatially explicit approach to global variance decomposition in MCDM was proposed by Ligmann-Zielinska and Jankowski (2014) by developing a procedure to calculate and map spatially distributed measures of model input uncertainty and model output sensitivity. Applications of this approach can be found in landslide hazard analysis (Feizizadeh et al. 2014) and land use change analysis (Şalap-Ayça et al. 2018).

Consequently, one can divide the applications of spatial US-A into two categories. The first category comprises models featuring spatial inputs and outputs, yet the US-A is conducted on variables representing aggregated spatial outputs. The second category includes models where spatial outputs are directly used in US-A, resulting in uncertainty and sensitivity maps. This approach is rare in spatial modeling. A notable early example includes a study by Herman et al. (2013a), who performed sensitivity analysis on a hydrological model performance metrics for each grid cell in a drainage river basin, effectively producing sensitivity maps. This approach offers the advantage of identifying patches of land where distinct input factors play a dominant role in

contributing to large streamflow events in the basin. Similarly, Ligmann-Zielinska (2013) used land-use change maps and an agent-based residential development model to identify the geographic locations where specific inputs predominantly drive development frequency. In addition, this study compared sensitivity maps with aggregate spatial statistics of development. The findings showed that the neighborhood effect can significantly influence outcome variance when development is considered as a whole. However, the neighborhood effect might not dominate the variability of development when analyzed through land use change maps. Finally, the study by Ligmann-Zielinska and Jankowski (2014) performed spatially explicit US-A to evaluate where specific input weights drive habitat suitability. In summary, the benefit of spatially explicit US-A is that, in addition to addressing the question of which model inputs have a dominant effect on outcome uncertainty, one can also investigate where this dominance occurs.

Due to the geographic nature of the output variables, a spatially explicit US-A poses unique challenges. The most prominent challenges pertain to computational cost, exemplified by the extended execution times of US-A, and the cognitive effort required to interpret results.

First, spatially distributed outputs often require long processing time and ample storage space, especially if the number of mapping units is large (Herman et al. 2013b). For stochastic spatial models, outputs can be easily counted in terabytes. The high computational cost of processing two-dimensional data, coupled with many model runs, requires efficient algorithms and parallelization. Various parallelization approaches have been proposed to reduce the computational cost (Kang et al. 2020). The computational time can also be reduced by using surrogate models (Şalap-Ayça et al. 2018) or applying a time-efficient screening before a more complex global analysis is performed (Herman et al. 2013b). The quality of analysis is critical for correctly identifying the drivers of model variability.

Second, a graphical presentation of analysis results can potentially hinder or support cognition and understanding depending on the effectiveness of the presentation. Since model spatial outputs are in the form of maps, it stands to reason that US-A results should also be presented as maps. Global US-A aims to elucidate model sensitivity by identifying influential inputs. Consequently, each input is assigned its own measure of contribution to the model's variability. The number of US-A maps to evaluate concurrently can easily exceed several dozens, hindering the interpretation of results. In addition, it is generally accepted that interpreting maps is cognitively more demanding than interpreting scalar values. Hence, reducing the number of US-A maps is even more critical. One solution is to concurrently visualize multiple sensitivity measures using methods such as bivariate choropleth classifications (Trumbo 1981), dominance maps (Ligmann-Zielinska and Jankowski 2014), or coincident maps (Şalap-Ayça et al. 2021).

In light of these challenges, we propose a two-stage spatially explicit global US-A aimed at screening the initial pool of inputs to eliminate non-influential ones, simplify a model, and reduce US-A computational cost. We then refocus our analysis on quantifying the influence of the remaining inputs on model output variability. This two-stage US-A comprises Morris's method described in Section 2.3.1 (Morris 2006), followed by

the quantification of Sobol sensitivity indices described in [Section 2.3.2](#) (Sobol 1993). Additionally, we propose bivariate mapping to visualize US-A results, reducing the number of maps requiring interpretation by half.

The proposed approach to US-A for models with spatially distributed outputs has two potential benefits. First, it reduces computational cost through model simplification. Second, it lowers the cognitive effort of interpreting US-A results through a compact visualization form offered by bivariate maps.

The joint use of the Morris and Sobol methods has become a common practice, especially in the last decade (Campolongo et al. 2011, Shin et al. 2013, Garcia et al. 2019, Nabi et al. 2021, Borgonovo and Rabitti 2023), including spatial modeling (Pappenberger et al. 2008, Herman et al. 2013b). For example, Ligmann-Zielinska and Jankowski (2014) used the Sobol method to produce uncertainty and sensitivity maps for habitat suitability. However, their approach did not use the Morris method as an introductory step to reduce the model input space. Additionally, they used individual sensitivity maps rather than the bivariate mapping proposed here. Only one study, conducted by Herman et al. (2013b), explored spatial outputs by jointly utilizing both the Morris and Sobol methods. However, their focus was on comparing the computational efficiency of both approaches rather than the use of the Morris method as an introductory stage to the Sobol method, which is one of the objectives addressed here.

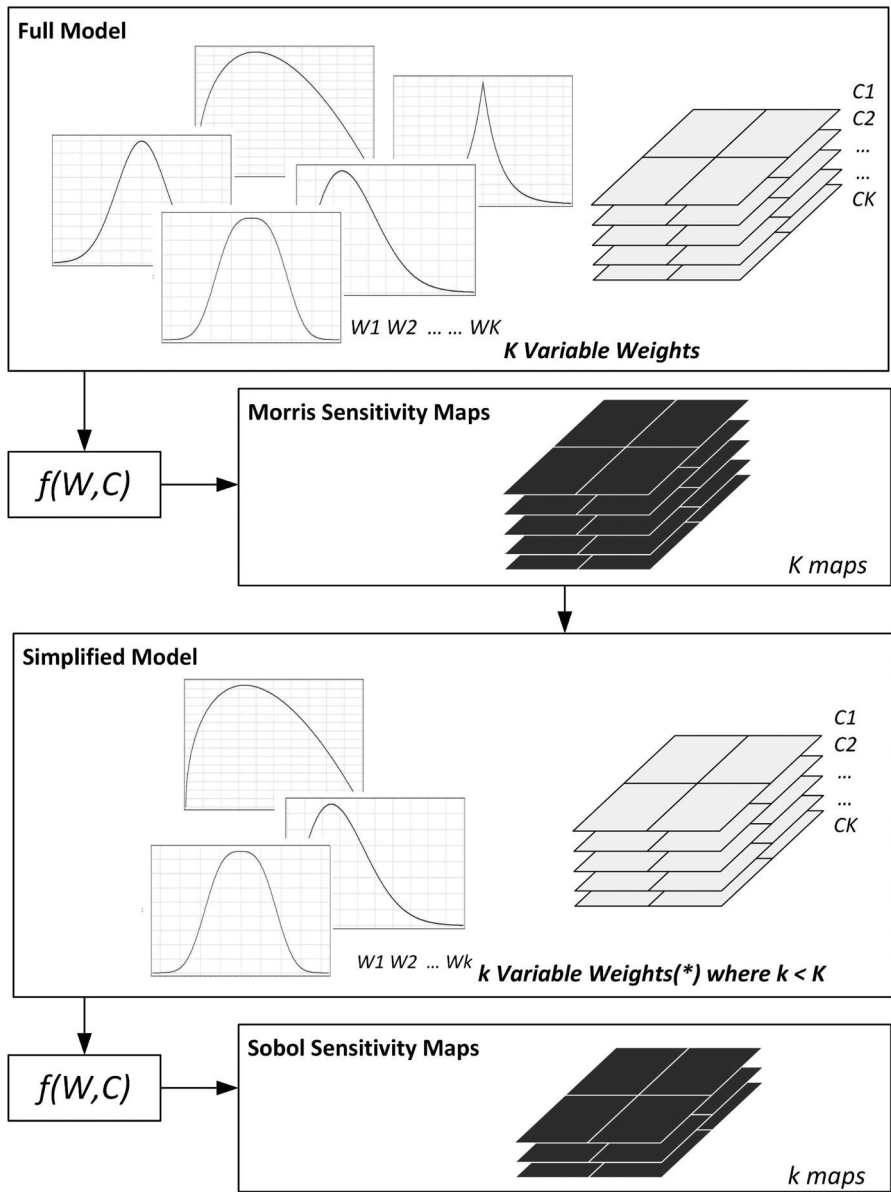
We demonstrate our framework by analyzing the uncertainty and sensitivity of an example MCDM called the geodiversity assessment. Geodiversity refers to the variability in abiotic natural components in a given area (Gray 2004, Zwoliński 2004). Among other inputs, these components include landforms, lithology, relief energy (ie relative height), soils, land cover/land use, surface and groundwater, and mesoclimate. Information on geodiversity can assist in managing protected and conserved areas, promote ecotourism, and support the evaluation of ecosystem services (Gray 2013).

The geodiversity assessment presented here calculates a composite indicator by aggregating spatial criteria (maps) with their aspatial importance measures (weights). In this study, only the weights are the variable inputs. The criteria maps remain constant across multiple executions. The Roztocze National Park (RNP), established in 1974 in southwestern Poland, was selected as the study area.

2. Framework

This section provides an overview of the framework used in the study. The procedure of the joint screening and variance decomposition of models with spatial outputs is presented in [Figure 1](#).

The screening method, also known as Morris, is initially applied to a model with K variable inputs. We refer to it as a full model since it involves all the initial inputs, which, in our case, are MCDM weights derived from expert opinion surveys ([Section 3](#)). This step involves fixing some of the non-influential inputs to constant values, resulting in a simplified model with fewer variable inputs, denoted as k . By non-influential, we mean weights that have no effect on the variability of the geodiversity score maps. Since they do not contribute to this variability, they may be set to constant values, reducing model input variables' space and producing a simplified model. The simplified



(*) for $K - k$ criteria, the weights are set to constant values

Figure 1. Framework of model simplification by input fixing based on spatial sensitivity analysis. The inputs include K variable aspatial weights W , and K constant spatial criteria maps C . After calculating the composite index using a function of C and W , K sensitivity maps are generated using the screening method (Morris method) – one map per weight. Screening allows for setting the non-influential inputs to constants, reducing the set of weights from K to k where $k < K$. The composite index is then recalculated using the reduced number of weights to generate sensitivity maps through variance decomposition (Sobol method), which are used to identify inputs driving output variability.

model is then utilized in the more computationally demanding sensitivity analysis based on the Sobol variance decomposition. It is important to note that both methods produce sensitivity maps in the proposed framework. To distinguish between the methods, we call these spatial outputs Morris maps and Sobol maps, respectively. Both methods are described in [Section 2.3](#).

2.1. Geodiversity assessment

In the research presented here, we used spatial multicriteria analysis to combine expert judgment with underlying abiotic data that define geodiversity (Sharples 1995, Australian Heritage 2002, Gray 2004, Zwoliński 2004). The data comprises criteria maps and their corresponding weights. Each spatial assessment unit (eg polygon or grid cell) is assigned a geodiversity assessment score. In reference to the contemporary understanding of geodiversity and following the geodiversity assessment (Zwoliński et al. 2018, Jankowski et al. 2020, Najwer et al. 2022, Najwer et al. 2023), seven abiotic components were taken into account: geomorphology (G) represented by types of landforms, hydrography as a topographic wetness index (H), lithological features (L), mesoclimate as total insolation (M), land cover and land use as an anthropopression indicator on the Earth's surface (P), relief energy represented by relative height (R), and types of soils (S):

$$GD = f(G_C, G_W, H_C, H_W, L_C, L_W, M_C, M_W, P_C, P_W, R_C, R_W, S_C, S_W) \quad (1)$$

where GD denotes standardized (0-100 range) geodiversity value, subscript C stands for criteria maps and subscript W denotes weights. Both criteria and weights are based on expert assessment ([Section 3](#)). All these inputs are susceptible to ambiguity, as expert judgments may differ among individuals, and the data may vary in quality. Accounting for these uncertainties improves the robustness of geodiversity assessment, which is achieved through US-A.

2.2. Calculating geodiversity

We compute the final geodiversity score using the aggregate function of the Technique for Order of Preference by Similarity to the Ideal Solution, commonly known as TOPSIS (Hwang and Yoon 1981). The advantage of TOPSIS over other MCDM techniques, such as AHP, lies in simplicity and direct ranking. TOPSIS is simpler to understand and implement compared to AHP, which involves pairwise comparisons increasing non-linearly in relation to the number of criteria. Moreover, TOPSIS provides a direct ranking of alternatives based on their overall similarity/dissimilarity to/from the ideal/nadir solution. This direct measurement avoids potential correlations between criteria in an MCDM model, which – if present, contributes to double-counting the same effect. Given the differences between TOPSIS and other MCDM techniques, it is likely that their results will differ.

TOPSIS accounts for potential dependencies between the criteria and is based on calculating distances between weighted criteria values and two extreme metrics, the ideal and the nadir. The ideal metric represents the best possible performance for each criterion. It is constructed by taking the maximum value for benefit criteria and

the minimum value for cost criteria across all alternatives. The nadir metric represents the worst performance for each criterion. It is constructed by taking the minimum value for benefit criteria and the maximum value for cost criteria across all alternatives. The spatial unit closest to the ideal and farthest from the nadir is assigned the highest geodiversity assessment score.

Specifically, based on our geodiversity assessment definition, we have seven criteria scores per each spatial assessment unit (su):

$$[G_C^{su}, H_C^{su}, L_C^{su}, M_C^{su}, P_C^{su}, R_C^{su}, S_C^{su}] \quad (2)$$

After applying the seven respective weights, we obtain the following values V :

$$[G_V^{su}, H_V^{su}, L_V^{su}, M_V^{su}, P_V^{su}, R_V^{su}, S_V^{su}] \quad (3)$$

where, for example:

$$G_V^{su} = G_C^{su} * G_W^{su} \quad (4)$$

Next, we identify sets V for the ideal and the nadir:

$$Id^* = [G_V^*, H_V^*, L_V^*, M_V^*, P_V^*, R_V^*, S_V^*] \quad (5)$$

$$Nd \sim = [G_V^{\sim}, H_V^{\sim}, L_V^{\sim}, M_V^{\sim}, P_V^{\sim}, R_V^{\sim}, S_V^{\sim}] \quad (6)$$

Where Id^* is a set of maximum V extracted from each weighted criteria layers, and $Nd \sim$ is a set of minimum V extracted from each of the seven weighted criteria layers.

Next, for every su , we calculate its separations from Id^* and $Nd \sim$, respectively:

$$Sep_{su}^* = \sqrt{(G_V^{su} - G_V^*)^2 + (H_V^{su} - H_V^*)^2 + \dots + (S_V^{su} - S_V^*)^2} \quad (7)$$

$$Sep_{su}^{\sim} = \sqrt{(G_V^{su} - G_V^{\sim})^2 + (H_V^{su} - H_V^{\sim})^2 + \dots + (S_V^{su} - S_V^{\sim})^2} \quad (8)$$

The final geodiversity (GD) assessment score is calculated as follows:

$$GD^{su} = \frac{Sep_{su}^{\sim}}{Sep_{su}^{\sim} + Sep_{su}^*} \quad (9)$$

where; $0 \leq GD^{su} \leq 1.0$

2.3. Sensitivity analysis measures

As already mentioned, the US-A employed in this research has two stages (Figure 1), producing different sensitivity measures. First, we apply screening to all weights ($K=7$) to weed out inputs that do not affect the geodiversity assessment score. Screening is done using two measures of sensitivity per input i – absolute mean (μ_i^*) and standard deviation (σ_i^2). Next, we apply variance decomposition on the subset of k remaining weights to evaluate the individual and joint influence of these weights on the final geodiversity assessment score. Variance decomposition results in two sensitivity indices per weight i – first order (S_i) and total effects (ST_i). Below, we provide an overview of the quantification of all four measures. More details can be found in the corresponding references.

2.3.1. Screening

The screening method was introduced by Morris (1991, 2006) and later revised by Campolongo et al. (2007). Screening (*aka* Morris) is frequently called an elementary effects method, based on multiple model executions of randomized OAT variations (Borgonovo and Rabitti 2023). Specifically, each input (i) is perturbed in the input space to create trajectories (j) that are used to estimate r number of elementary effects (EE_i^j), which are simply ratios of the changes in model output to the changes in i . These elementary effects constitute a distribution (Saltelli et al. 2008, pp. 110–121).

The major advantage of Morris method, compared with other methods, stems from its low computational cost as it does not require many model executions (Campolongo et al. 2007, Herman et al. 2013b). The outcome of screening is a set of measures per input, of which two are of particular interest – an absolute mean (μ^*), which measures the overall impact of an input on the model result (both first and higher orders), and a standard deviation (σ), which encompasses interactions of a given input with other inputs (Campolongo et al. 2007):

$$\mu_i^* = \frac{1}{r} \sum_{j=1}^r |EE_i^j| \quad (10)$$

$$\sigma_i^2 = \frac{1}{r-1} \sum_{j=1}^r (EE_i^j - \mu)^2 \quad (11)$$

Note that, for models producing spatial outputs, these two metrics are calculated independently per each spatial unit (Herman et al. 2013a) producing sensitivity maps. For example, with five inputs and ten watersheds, we have 100 metrics to analyze.

While these measures do not carry any direct information on the quantitative contribution of inputs to output variability (Campolongo et al. 2007), they allow for prioritizing the inputs by ordering them based on the value of μ^* , where higher values indicate influential inputs and lower values point to inputs that are negligible and can be set to constant values, reducing model input space. Furthermore, the relatively high values of σ are indicative of inputs that are involved in interactions. The Morris method identifies the most sensitive inputs, which can be further examined in other global sensitivity analyses like variance decomposition.

2.3.2. Variance decomposition

The geodiversity assessment realizations constitute a two-dimensional distribution, which can be summarized by calculating, for each spatial unit, the standard descriptive statistics like mean or variance. Both metrics can be used to generate uncertainty maps (Section 4.2). The variance can be then decomposed to produce partial variances, which inform us about input influence on output variability:

$$V = \sum_i V_i + \sum_{i < j} V_{ij} + \sum_{i < j < m} V_{ijm} + \dots + V_{12\dots l} \quad (12)$$

Where V is total variance, V_i is partial variance used to calculate model sensitivity only due to input i , V_{ij} represents sensitivity due to the interaction between inputs i

and j , V_{ijm} represents sensitivity due to interactions among i , j , and m , and $V_{12...l}$ signifies model sensitivity due to interactions among all l inputs.

This procedure produces two sensitivity indices per input (*aka* Sobol indices): first order index S_i and total effects index ST_i (Sobol 1993, Sobol 2001, Homma and Saltelli 1996, Saltelli et al. 2008):

$$S_i = \frac{V_i}{V} = \frac{V_{X_i}[E_{X_{-i}}(Y|X_i)]}{V(Y)} \quad (13)$$

$$ST_i = \frac{V(Y) - V_{X_{-i}}[E_{X_i}(Y|X_{-i})]}{V(Y)} = S_i + S_{ij} + S_{im} + \dots + S_{ijm} + \dots + S_{ij...l} \quad (14)$$

The S_i index informs about the input's contribution to the total variance of model output, taken independently from the $l - 1$ inputs. ST_i , on the other hand, captures the input's overall contribution to output variability – both singly and in interactions with other inputs, where $V_{X_{-i}}[E_{X_i}(Y|X_{-i})]$ is the total contribution of all non- i inputs to the variance of Y so that $V(Y) - V_{X_{-i}}[E_{X_i}(Y|X_{-i})]$ captures the remaining conditional variance due to i .

The major advantage of Sobol's approach is its model independence – it can be used for both simple (linear, additive) and more complex (nonlinear) models (Saltelli et al. 2008). It is, therefore, applicable to other spatial models apart from MCDM. It also provides two succinct measures of input influence that can be used, among other things, for input prioritization for further research, input fixing for model simplification, or model consensus building and policy analysis (Tarantola et al. 2002, Saltelli et al. 2008, Mara et al. 2017, Razavi et al. 2021).

Similarly to Morris, given the spatial nature of our outputs, both the S_i and ST_i indices are computed independently for each spatial unit. This process effectively generates sensitivity maps (Herman et al. 2013a, 2013b, Ligmann-Zielinska and Jankowski 2014).

In summary, the Sobol method provides insights into inputs that individually contribute to output variance, inputs involved in interactions, and inputs with both individual and interactive contributions to output variance.

2.4. Bivariate mapping

Campolongo et al. (2007) demonstrated that both μ^* (Morris) and ST (Sobol) could be jointly used to qualitatively assess model sensitivity. The metrics provide similar information about the importance of inputs, even though they are formulated differently from a statistical standpoint.

Both Morris and Sobol's methods produce two sensitivity measures per input. Consequently, to gain complete insight into spatial model sensitivity, we should evaluate two types of maps. For example, our case study would amount to 14 Morris maps and 14 Sobol maps (Figure 1). Such an approach is cognitively demanding and may preclude deriving any meaningful conclusions from the analysis. Instead, we propose rendering both metrics on a single map using bivariate choropleth classification to reduce the number of global sensitivity analysis outputs to analyze. A bivariate map offers the ability to visualize the relationship between two different variables

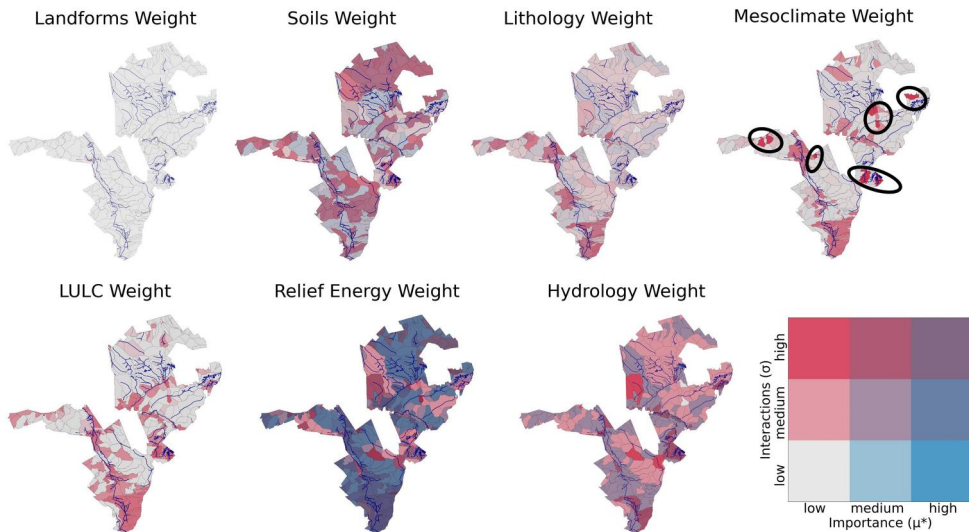


Figure 2. Bivariate screening sensitivity maps, generated using the Morris method for seven weights. The outlined watersheds for the mesoclimate weight have considerably higher bivariate scores than their neighborhoods.

simultaneously, which a single-variable map does not provide. By displaying two variables in a single map, viewers can discern potential correlations, patterns, or trends that may not be apparent when looking at each variable in isolation.

For Morris maps, in the bivariate mapping, the horizontal axis depicts μ^* and the vertical σ (Figure 2, legend). Correspondingly, for Sobol maps, the horizontal axis denotes the ST_i index, while the vertical represents the $(ST_i \text{ minus } S_i)$ index, signifying weight interactions (Figure 3, legend). In both cases, the horizontal axis informs about the overall effect of weight on geodiversity score variability, whereas the vertical axis provides additional information about weight interactions.

While it would be sufficient to compare Morris maps of μ^* with Sobol maps of ST (Herman et al. 2013b), adding the dimension of the interactions enables the identification of regions within the maps where the influence of various inputs becomes more complex due to their interactions. Such information may be helpful for input fixing or explanation of dynamic model behavior.

3. Data and experimental design

The RNP is located in the upland landscape zone of south-eastern Poland, covering an area of 8482.83 ha, with 92% of it covered by forest, which is predominantly strictly protected. Quaternary sediments on the surface consist of slope formations (2250 ha), aeolian and fluvial sands (1000 ha), loess, and occasional glacial, lake and swamp deposits. Accumulation forms encompass dunes, dune chains, deflation plains, and basins. Drainless depressions are filled with marshy accumulations, such as peats and mineral-organic silts. Noteworthy are loess covers, within which erosional forms have developed, often in the form of gullies. The landforms are characterized by suffusion

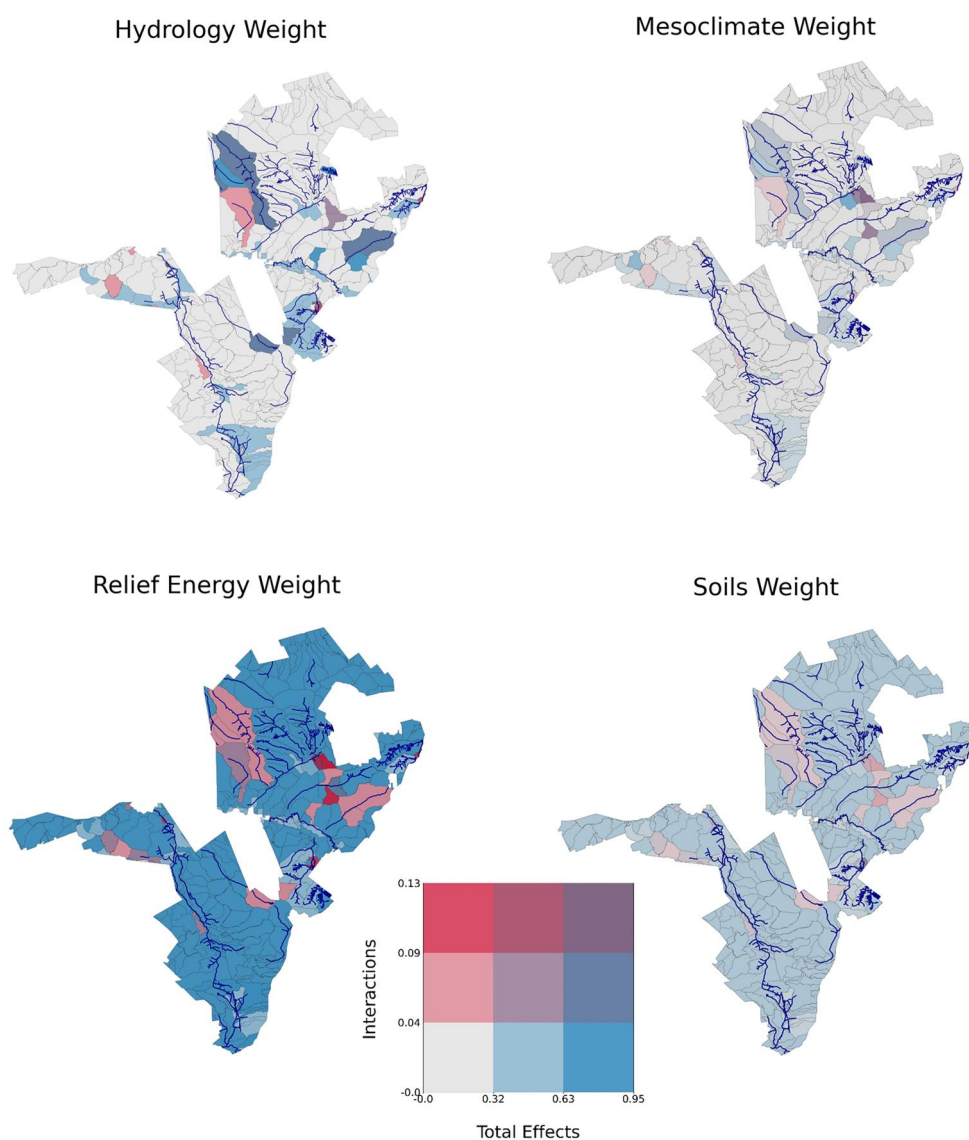


Figure 3. Bivariate sensitivity maps generated from variance decomposition (Sobol) illustrate the overall contribution of weight factors to the total variance on the horizontal axis, while representing interactions between weight factors on the vertical axis.

holes in the upper part and active alluvial cones at their outlets. The RNP area includes a total of 403 designated catchments.

This research used the geodiversity assessment computed for each watershed and described in Jankowski et al. (2020) and Najwer et al. (2022). Details on the input source datasets for the computation are available in the ‘readme’ file, accessible through the accompanying code link.

Both weight and criteria values were obtained through an expert survey. The criteria maps reflect each criterion’s assessed/judged contribution to geodiversity rather

Table 1. Weight definitions used in sampling.

Name	Symbol	Weight Values (%)	Probabilities
Landforms Weight	G _W	4, 10, 12, 13, 15, 16, 17, 19, 20, 25, 26, 27, 30, 31, 40	0.02, 0.02, 0.02, 0.02, 0.18, 0.04, 0.04, 0.02, 0.32 , 0.12, 0.02, 0.02, 0.14, 0.02, 0.04
Hydrology Weight	H _W	5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 17, 19, 20	0.14, 0.02, 0.04, 0.02, 0.04, 0.3, 0.04, 0.02, 0.07, 0.16 , 0.02, 0.02, 0.14
Lithology Weight	L _W	7, 10, 12, 14, 15, 16, 17, 19, 20, 25, 26, 27, 28, 30, 35, 40, 45	0.02, 0.07, 0.02, 0.02, 0.23 , 0.02, 0.05, 0.02, 0.3, 0.12, 0.04, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02
Mesoclimate Weight	M _W	1, 2, 3, 5, 6, 9, 10, 11, 12, 13, 14, 19, 20, 25	0.04, 0.04, 0.04, 0.37 , 0.04, 0.04, 0.25, 0.02, 0.04, 0.04, 0.07, 0.02, 0.02, 0.02
Land use/land cover Weight	P _W	1, 2, 3, 4, 5, 6, 10, 12, 13, 14, 15, 17, 18, 20	0.02, 0.02, 0.02, 0.02, 0.25 , 0.04, 0.18, 0.04, 0.02, 0.07, 0.18, 0.04, 0.02, 0.12
Relief Energy Weight	R _W	5, 8, 10, 11, 13, 14, 15, 17, 18, 19, 20, 21, 22, 25, 30, 35	0.05, 0.02, 0.19 , 0.02, 0.05, 0.04, 0.16, 0.04, 0.04, 0.02, 0.19 , 0.02, 0.02, 0.09, 0.05, 0.02
Soils Weight	S _W	1, 5, 6, 8, 9, 10, 11, 12, 14, 15, 17, 19, 20, 25	0.02, 0.18, 0.04, 0.05, 0.02, 0.28 , 0.04, 0.02, 0.05, 0.19, 0.02, 0.02, 0.07, 0.02

We used raw weight values, ie the complete enumeration of values assigned by the respondents and their corresponding frequencies. Bolded weights have the highest probability of selection.

than the underlying raw abiotic data (like soil types, landforms, or land use). Hence, all criteria share the same value domain, ie a five-point Likert scale where one is the lowest, and five is the highest contribution of criterion to geodiversity. For this study, we only kept the weights as variable inputs (Table 1), and for the criteria, we mapped average criterion values based on all 33 respondent inputs. These averaged criteria maps are used as constant inputs in the assessment and are summarized in Figure 4. For example, on average, the respondents assigned relatively high ratings to mesoclimate regardless of the watershed, producing a relatively homogeneous map. Conversely, watersheds relative ratings of the relief energy criteria are spatially heterogeneous, with a clear cluster of high values in the central pass connecting the southern and the northern RNP regions.

The criteria weights were elicited through the geo-questionnaire method described in Jankowski et al. (2020). The weights were specified by experts in Earth Sciences recruited from 24 countries. They reflected individual expert judgments and introduced an obvious uncertainty bias into the geodiversity assessment, which is the focus of the analysis presented here.

3.1. Sampling and execution

Morris and Sobol’s methods require different sampling methods. This research used SALib (Herman and Usher 2017) implementation of Morris with default input values. Hence, for Morris, the total number of model executions amounted to the following:

$$R = (k + 1) * N \tag{15}$$

where *N* is the user-defined number of runs, and *k* is the number of inputs. We set *N* to 2000 in the reported analysis to maximize index accuracy.

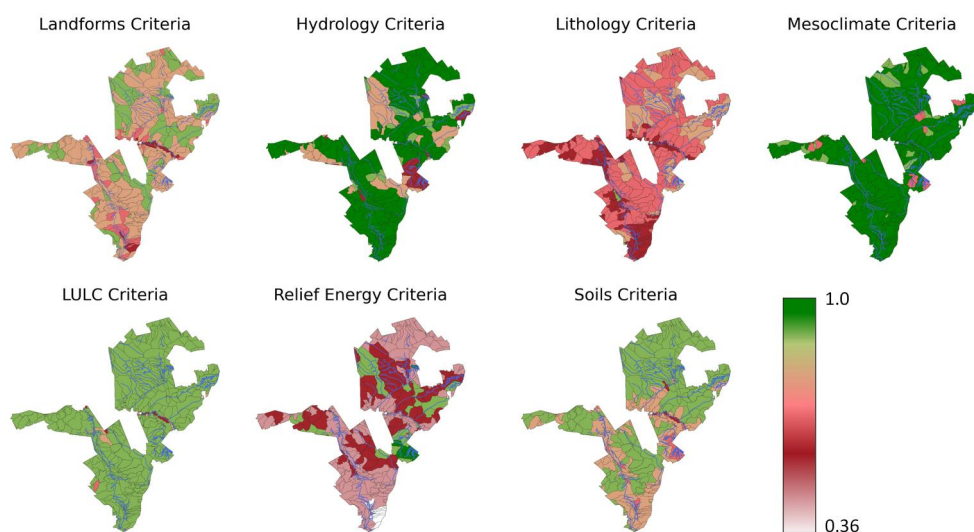


Figure 4. Standardized average values of the input criteria maps for the RNP study area, obtained from 33 respondents. These average criteria maps were directly used in geodiversity assessment calculations. High values indicate regions where most respondents identified a given criterion as particularly important for geodiversity assessment.

We used the Sobol quasi-random sequence to calculate S and ST , improving samples' uniformity in the input space (Sobol 1993). Further, we applied the radial version of the quasi-random sampling for Sobol indices, introduced by Saltelli et al. (2010), which has proven to produce optimal results with fewer runs. The fastest convergence of the indices (due to improved uniformity of samples) is obtained when the number of runs is the power of two (Owen 2020). Consequently, the formula for the number of runs for Sobol quasi-random sampling is:

$$R = p^2(k + 2) \quad (16)$$

Where p^2 reflects the number of runs (aka N), k is the number of variable inputs, and $(k + 2)$ are the radial combinations (Saltelli et al. 2010). To ensure that the indices converge to acceptable values (ie $Si \leq STi$ and $Si \geq 0$), we set $p = 17$.

The execution time for Morris was 57 seconds, whereas it took 8 minutes to execute Sobol. We used 64-bit OS with Intel® Core™ i7-8550U CPU @ 1.8 GHz, 1.99 GHz, and 16 GB RM for all simulations.

4. Results

We present the results of the US-A analysis in three sequential steps. First, we discuss the Morris maps showcasing which weights are designated as constants and which are preserved for variance decomposition, thereby streamlining the geodiversity assessment process. Subsequently, we touch upon the findings of the uncertainty analysis following the application of the Morris method. Finally, we transition to the Sobol

maps, highlighting the efficacy of bivariate mapping in condensing the multitude of maps for interpretation.

4.1. Screening

Morris sensitivity maps are presented in [Figure 2](#). The horizontal axis of the classification is the μ^* , which we refer to as the importance measure, whereas the vertical axis represents σ , called interactions. Since the raw values of both statistics are not a direct measure of input sensitivity, we decided to streamline the numeric classification into three ordinal values: low, medium, and high.

At first glance, the maps fall into two distinct categories: (a) maps where the joint classification shows relatively low bivariate values across the area (landforms, lithology, and land use/land cover) and (b) maps with low to high scores with a considerable spatial heterogeneity (hydrology, mesoclimate, soils, and relief energy). This binary categorization is crucial because it determines whether a criterion weight should be discarded in subsequent analysis. For example, consider the mesoclimate map. In certain regions (outlined in black), it scores very high on both the importance and the interactions; however, most watersheds have very low values (colored light gray). Given the former, we decided not to set this weight to a constant value.

The second kind of observation relates to the relative comparison of Morris maps. Certain weights, such as landforms, lithology, and land use/land cover, exhibit consistently low importance values across the entire study area and can be easily discarded. Additionally, the soils weight, characterized by low importance but medium interactions throughout the area, could also be excluded. However, we have chosen to retain the soils weight as a test case for comparison with Sobol sensitivity maps, as presented in [Section 4.3](#).

In short, for the Sobol analysis, four weights were kept as variable inputs (hydrology, mesoclimate, relief energy, and soils) and three weights – landforms, lithology, and land use/land cover – were set to constant values with the highest probability of selection.

4.2. Uncertainty analysis

[Figure 5](#) (top) is a graphic summary of the average geodiversity score before and after simplification, ie seven weights versus four weights. Note that the results are almost identical, which bodes well for the methodology since Morris should not substantially affect the results compared to Sobol. [Figure 5](#) bottom shows the uncertainty maps for both experiments. These maps allow for identifying regions that require more attention when investigating the sensitivity analysis scores. First, we can identify regions of low versus high values of the average geodiversity assessment score. Second, we can simultaneously evaluate where geodiversity score variability (standard deviation) is the highest. The regions of higher uncertainty are of particular interest, as they provide the basis for variance decomposition (trivially, if the standard deviation and hence variance are low, there is little to decompose).

We can observe minor differences in the uncertainty maps between the full and the simplified geodiversity assessment, that is, the standard deviation does not

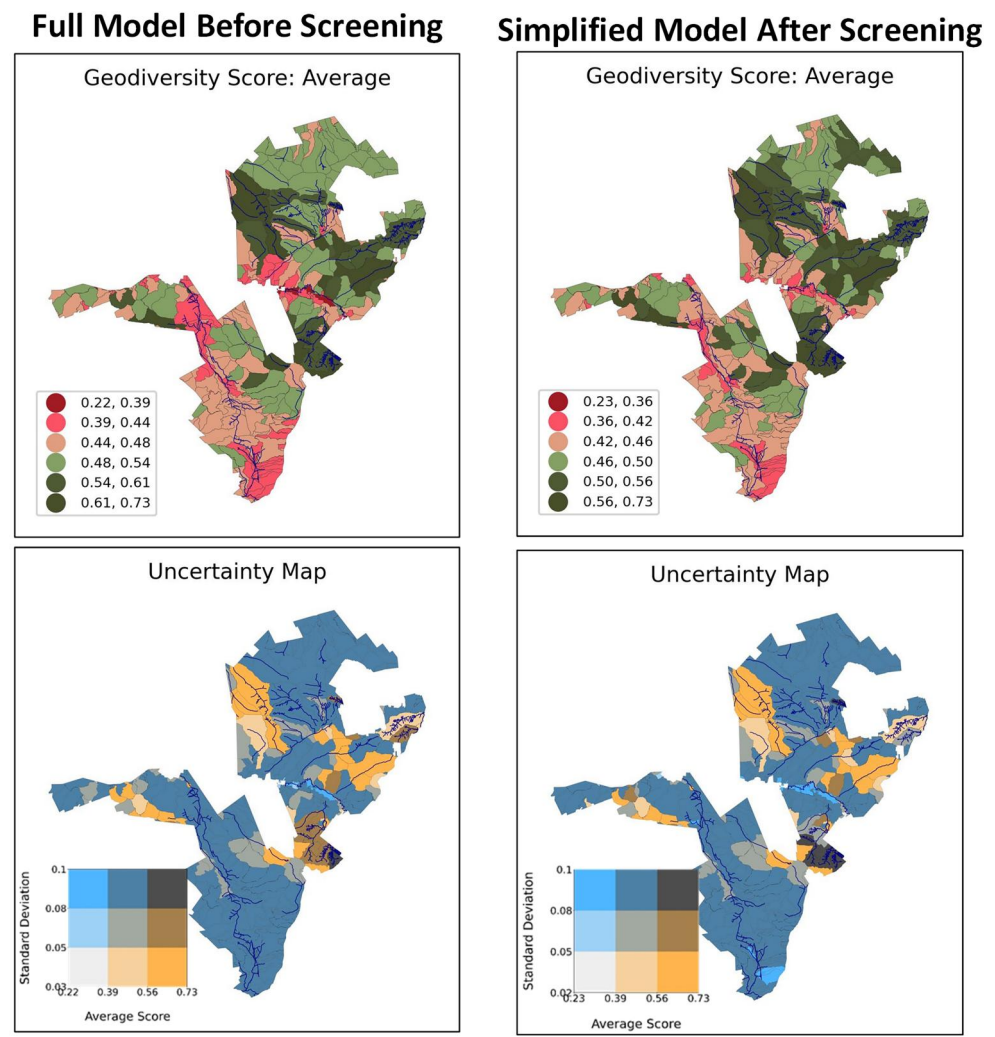


Figure 5. Comparison of the comprehensive geodiversity assessment utilizing seven weights against the simplified version employing four weights: average geodiversity maps (above) juxtaposed with their corresponding bivariate uncertainty maps (below).

considerably vary between the two experiments. Therefore, upon close examination of the classification, it becomes evident that the majority of watersheds either remain within their current class or transition up or down by one class. Overall, the simplification proved to maintain the general trends in the results, as should be expected.

4.3. Variance-based sensitivity analysis

The Sobol experiments result in computing two sensitivity indices per input. In our case, the most intuitive approach would be to map each of these indices separately, ie generate four S_i and four ST_i maps. Figure 6 shows the corresponding maps for the

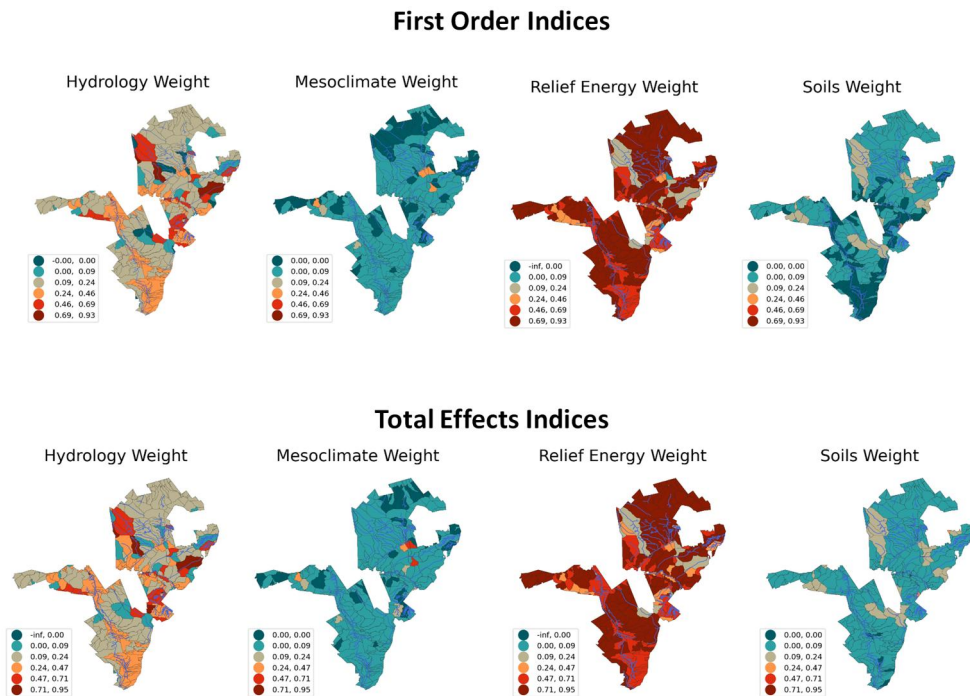


Figure 6. S and ST sensitivity maps obtained from variance decomposition (Sobol method) for the four variable weights used in the simplified geodiversity assessment.

four criteria. Note that there is minimal difference between the first order and total effect maps due to the generally low standard deviation of the geodiversity score.

4.3.1. Bivariate mapping

To simplify the interpretation of the Sobol maps, we combined the mapping of ST_i and interactions, as illustrated in Figure 3. This approach effectively reduces the sensitivity maps by half compared to Figure 6.

While the bivariate classification of Morris maps (Figure 2) and Sobol maps (Figure 3) for the four remaining weights is not identical, we can still observe certain similarities and highlight differences. First, in both cases, the relief energy weight drives the variability of the geodiversity score the most. Second, the mesoclimate weight, which scored the lowest out of the four after the Morris analysis, maintained its lowest rank when mapped using Sobol indices.

The most noticeable differences between both methods include the diminished influence of the hydrology weight, which, in Sobol maps, became more spatially heterogeneous, with a few watersheds of high influence surrounded by many watersheds with low to none influence. Moreover, there's a general decrease in the number of watersheds classified as 'high overall effect and high interactions,' referring to the upper right class in the bivariate classification.

In summary, Morris's maps and Sobol's maps show certain overlaps but also considerable differences. Possible reasons for these discrepancies can be attributed to methodological differences between the statistics, the relatively low variance across the

area, and the cartographic classification. First, the equivalence of μ^* and ST_i may be more complex for spatial outputs, especially when there is a difference in the number of weights to compare (seven versus four). While both statistics are strongly correlated, discrepancies may occur, as pointed out by Campolongo and Saltelli (1997).

The variable number of weights may affect the sampling and the computation of output statistics. Also, the differences may be challenging to identify when the geodiversity score is relatively stable across simulations. Furthermore, we used the natural breaks (Jenks 1967) method in all cases to divide the bivariate data into nine classes. While having three classes per dimension might seem relatively limited, given the two-dimensional nature of our representation, we made a deliberate choice to prioritize facilitating comparative analysis across multiple maps over intricate value detail. It's important to note that crisp boundaries between a few classes may result in less accurate conclusions about spatial patterns, and the observations must be evaluated with caution.

5. Discussion

Several research findings warrant further discussion, including implications for our case study, the advantages and disadvantages of using Morris and Sobol methods in tandem, the bivariate classification schemes, and potential directions for future research arising from the study's limitations.

First, the results have implications for our case study. Regardless of the differences in statistical values between Sobol and Morris's methods, we can identify two influential weights: relief energy and hydrology. The importance of these weights can be attributed to how respondents evaluated the abiotic characteristics. Relief energy plays an important role in geodiversity assessments. It is a morphometric determinant of relief and may indicate the pace and intensity of environmental changes. The relief energy reflects the rhythm of the landforms, indicating the fine- and large-radial diversity of the Earth's surface. Furthermore, relief energy contains information about the impact of gravity, which, alongside the morphogenetic input represented by water (reflected in the hydrology weight), plays an essential role in transforming morphological landscapes. In sum, these two inputs carry a significant amount of information about the morphogenesis, evolution, and current state of any area undergoing geodiversity assessment. They should always be considered when assessing the geodiversity of any area, especially protected and conserved ones.

While these observations hold practical value, it is essential to acknowledge that the data produced relatively stable results with low standard deviation, complicating variance decomposition. Overall, however, the results of Morris and Sobol's methods overlap and hence the geodiversity score calculated with four variable weights may be deemed acceptable.

Furthermore, considering that the underlying mathematical formalisms of Morris and Sobol's methods differ, differences in US-A results are expected. Both methods may yield similar sets of influential weights. However, input rankings based on sensitivities are likely to differ (Campolongo and Saltelli 1997, Herman et al. 2013b, Nabi et al. 2021). Sobol is generally recognized as a more precise approach simply because

it allows for numerical evaluation of the influence of the variable inputs. In contrast, the Morris method allows for ordinal comparison, primarily serving as a quick but approximate method to identify a subset of variable inputs for a more detailed quantitative Sobol analysis. The decision on which inputs to include and which to discard based on the ranking is relatively subjective, with no formal rules for identifying the cutoff values. The main advantage of using the Morris method as a preliminary stage to variance decomposition is reducing computational cost (Herman et al. 2013b), which enhances an operational use of the presented approach and improves its energy efficiency. Interestingly, this aspect of geocomputing has recently received attention, especially in light of Geo-AI, where massive computations are involved, generating a significant carbon footprint (Shi et al. 2023).

In Section 4.3, our primary focus was on comparing the results of both methods. It must be noted, however, that the methods are usually complementary rather than comparative (Garcia et al. 2019). Furthermore, Sobol and Morris's methods are not equivalent and are heavily dependent on sampling (randomization). Therefore, future studies should explore sampling schemes aimed at converging the results of both methods (Borgonovo and Rabitti 2023).

Moreover, these methods may not be suitable for situations involving non-normal distributions, as variance might not adequately summarize model variability. Note that when performing US-A, each spatial unit has its own distribution of results. Hence, interpreting US-A maps is akin to deriving conclusions on input importance across multiple aspatial variables. Chances are that many spatial units will have non-normal distributions. Future research will need to explore other US-A methods for spatial outputs that are not dependent on variance. One example is density-based, moment-independent measures (Pianosi and Wagener 2015).

Finally, bivariate classification choropleth maps may be more challenging to interpret than the more conventional single-variable choropleth maps (Reimer 2011). Therefore, in our future research, we plan to compare the efficacy of individual sensitivity index maps with bivariate maps using surveys.

6. Summary

The paper addresses two critical aspects of spatially explicit uncertainty and sensitivity analysis (US-A). Firstly, we introduce an innovative spatially explicit US-A framework that incorporates two sensitivity analysis methods applied to spatial outputs (maps) rather than their aggregates. These methods are applied sequentially, aiming to reduce the computational complexity associated with models featuring spatial outputs. While the joint use of Morris and Sobol sensitivity analysis methods is not unprecedented, their joint application for an in-depth analysis of spatial outputs is a unique contribution of this study.

Furthermore, we demonstrate that the computational and cognitive efforts required to interpret the more comprehensive Sobol global sensitivity analysis results can be reduced by removing non-influential inputs identified through the less computationally expensive Morris method. The screening process was aided by introducing bivariate maps, enhancing the visualization of input importance and interaction measures.

The framework has been exemplified in a geodiversity assessment case study, where four of the original seven inputs (weights) were recognized as non-influential and consequently excluded from subsequent analysis. Nevertheless, given their model-independent design, as extensively documented, both the Morris and Sobol methods hold applicability beyond the scope of multicriteria geodiversity assessment discussed in this study, extending to any model featuring spatially explicit input and output.

In conclusion, our approach directly addresses the three challenges outlined in the introduction. Firstly, the screening process applied to an initial geodiversity assessment with all variable inputs allows for a reduction in both computational cost and storage space. Secondly, global approaches are applied, resulting in a more thorough and accurate evaluation of input influence. Thirdly, bivariate choropleth mapping can alleviate the cognitive burden in deriving conclusions on the relative importance of inputs on spatial outputs.

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Authors contributions

All authors conceived and developed the idea and co-wrote the manuscript. Ligmann-Zielinska was responsible for code development, analysis, and article preparation. Jankowski refined the idea. Najwer and Zwoliński prepared the data, revised the application, and interpreted the findings.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data and codes availability statement

The code and data used in this study are available at the FigShare link: <https://doi.org/10.6084/m9.figshare.22811906.v3>. The link includes a 'readme' document with instructions on how to run the code and reproduce the results presented herein. The geodiversity assessment and US-A tools were developed using the Python Programming Language, version 3.10.5 (<https://www.python.org/>), packaged by conda-forge (<https://www.anaconda.com/>). Minimum software requirements, including third-party libraries with links, are listed in a table at the top of the 'readme' file. The authors of these libraries are not affiliated with this research.

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